

Home Assistant - Monthly Energy and Automation Report

Circle C Ranch residence - 60+ devices - Raspberry Pi 5 cluster (3 nodes)

Reporting period: February 2026

Energy Summary

Metric	Value
Solar generated	910 kWh
Home consumed	1,300 kWh
Powerwall cycled	absorbed 210 kWh, discharged 198 kWh
Grid import	470 kWh
Self-supply	64 percent

Automation Highlights

Circadian lighting (Lutron Caseta) held warm color temperature after sunset across all living zones. HVAC pre-cool ran on 11 mornings ahead of peak rates. The new presence-based room automation using mmWave sensors is in tuning, with the false-negative rate down to 3 percent after threshold adjustments.

Open Items

- Finish Powerwall reserve-tuning for winter peak shaving.
- Reduce mmWave dead-zone near the listening room reading chair.
- Add low-battery alerts for the 12 Aqara sensors past 20 percent.

Consumption by Zone

Zone metering is derived from circuit-level sensors and smart plugs reporting into Home Assistant. HVAC remains the largest single load through the cool season, ahead of the home lab rack.

Zone	Consumed (kWh)	Share
HVAC (two zones)	470	36 percent
Home lab and server rack	250	19 percent
Kitchen and appliances	210	16 percent
Living and bedrooms	185	14 percent
Nathan's piano studio	95	7 percent
Everything else	90	8 percent

Device Health

Category	Online	Attention
Lighting (Lutron Caseta)	18	0
Climate (Ecobee)	2	0
Cameras (Ring)	4	0
Presence and motion (Aqara)	14	3 low battery
Energy (Powerwall, inverter)	2	1 tuning

Automations Run This Period

Automation	Triggers	Result
Circadian lighting	62	Held warm evenings
HVAC pre-cool	11	Ahead of peak window
Presence room control	540	Tuning, 3 percent misses
Storm Watch charge	2	Pack topped ahead of weather
Low-battery notify	3	Flagged Aqara sensors

Six-Month Trend

Self-supply tracks solar season. Winter months lean on grid import while the shorter days limit generation. The rolling target remains about 75 percent across a full year.

Month	Generated (kWh)	Consumed (kWh)	Grid import (kWh)	Self-supply
2025-09	1,180	1,490	360	76 percent
2025-10	1,040	1,380	395	71 percent
2025-11	820	1,320	540	59 percent
2025-12	690	1,410	760	46 percent
2026-01	740	1,450	745	49 percent
2026-02	910	1,300	470	64 percent

Cost and Savings

Time-based control shifted evening load off the 4 PM to 9 PM peak window on most days. Estimated savings against a flat import scenario for the month is about 96 dollars, before the standing connection charge.

Next Period Focus

- Complete the seasonal reserve schedule so the pack drops to 20 percent when AC season begins in the spring.
- Replace batteries in the three flagged Aqara sensors.
- Extend presence coverage near the listening room reading chair.

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